



Getting Started with the ECU

1. Download and install the STM32 Virtual COM Port (VCP) drivers:

a. Option 1:

From the official STMicroelectronics website (registration may be required):

<https://www.st.com/en/development-tools/stsw-stm32102.html>

Get Software

Part Number	General Description	Latest version	ECCN (EU)	ECCN (US)	Supplier	Download
STSW-STM32102	STM32 Virtual COM Port Driver	1.5.0	NEC	3D991	ST	Get latest

b. Option 2:

From the RM Control Systems Google Drive:

<https://drive.google.com/file/d/1kvkKc6qwiY2arnYQB2QzxJUCPKBwo-Ut/view?usp=sharing>

2. Download and install tuning software:

a. Option 1: Download and install TunerStudio MS from EFI Analytics.

This guide will focus on using TunerStudio.

<https://www.efianalytics.com/TunerStudio/download/>



Note: There is a free version of TunerStudio with limited capabilities, or one-time pay registered versions with greater capabilities. The free version is sufficient for basic tuning, while the registered version unlocks advanced features such as VE Analyze Live, enhanced data logging, and additional tuning tools. Registered TunerStudio is recommended.

b. Option 2: Download and install EpicTuner.

EpicTuner is a free alternative to TunerStudio.

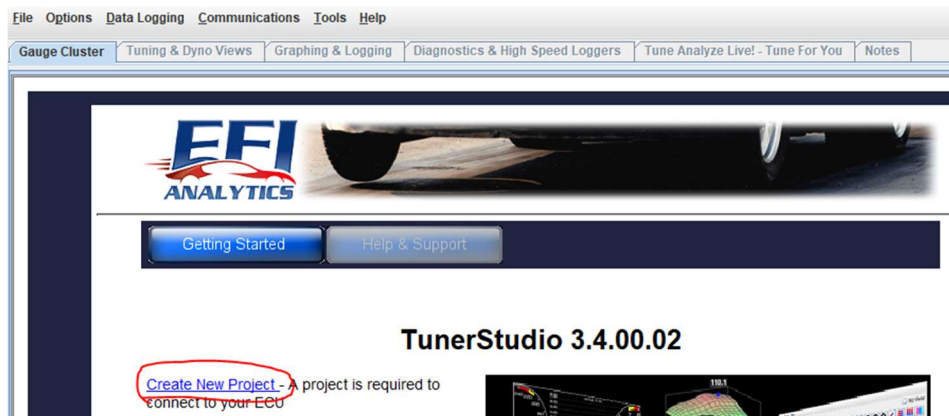
https://github.com/AKCore/epictuner_public/releases



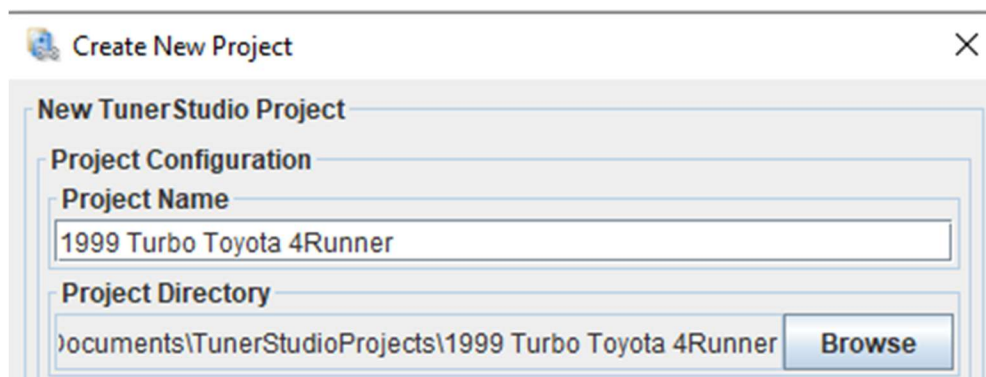
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3. Create a New Project in TunerStudio:

- a. **Important:** Many USB-C cables are charging-only and cannot transfer data. Make sure you have a USB-C cable that supports data transfer (not just charging!)
- b. Plug your laptop into the ECU via the USB-C cable. Your laptop should make an audible noise acknowledging that it sees the ECU.
- c. In TunerStudio, select “Create New Project”.



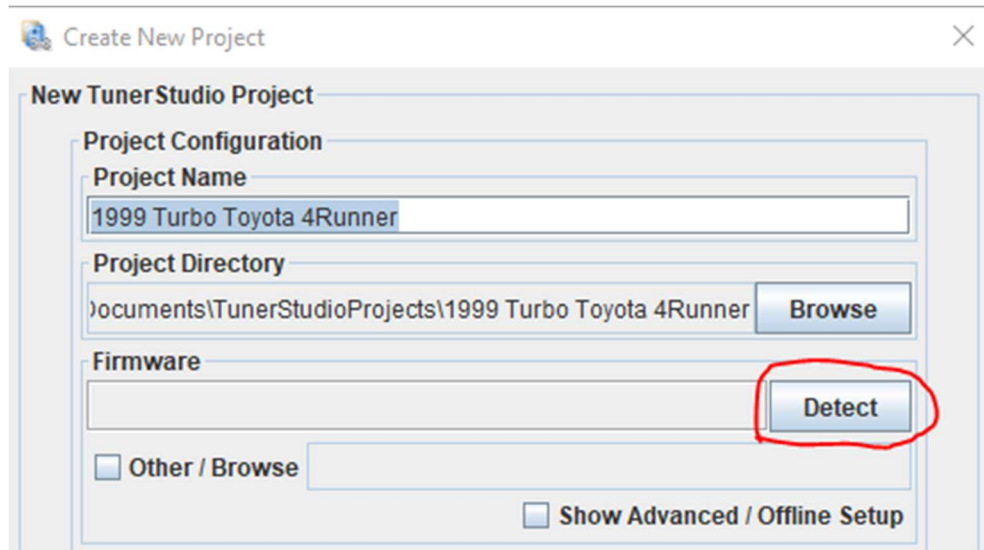
- d. Under “Project Name” choose a name for your project.
- e. Under “Project Directory” choose a location to save your project files (the default location is typically fine).



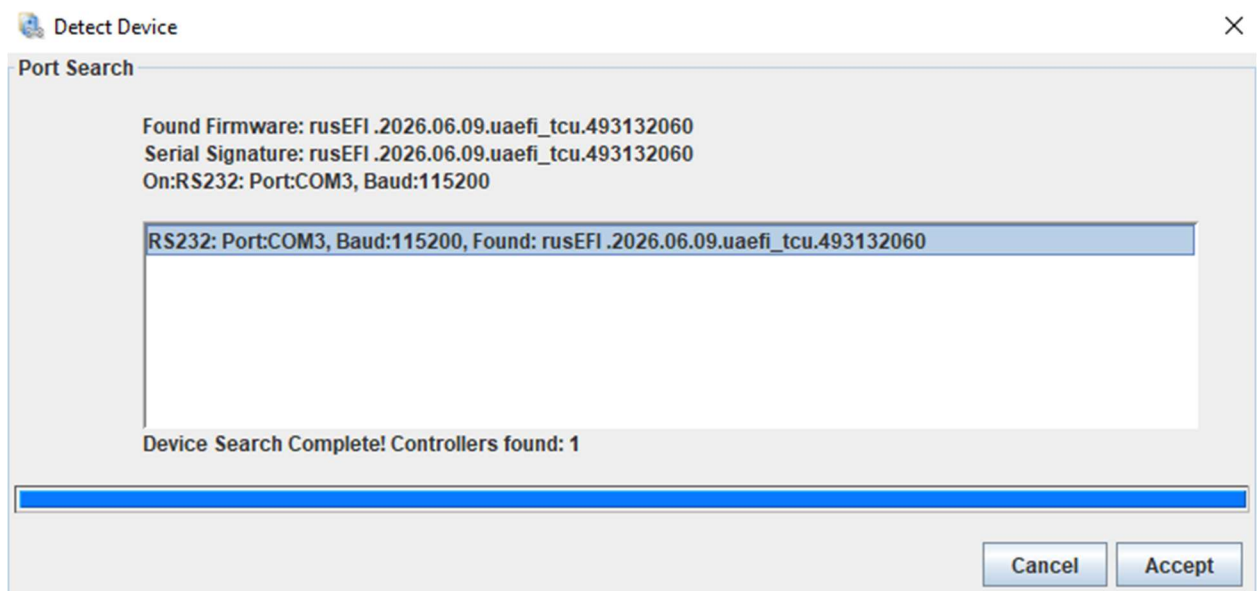


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f. Under “Firmware” select “Detect”.



g. Select the firmware detected and press “Accept”.





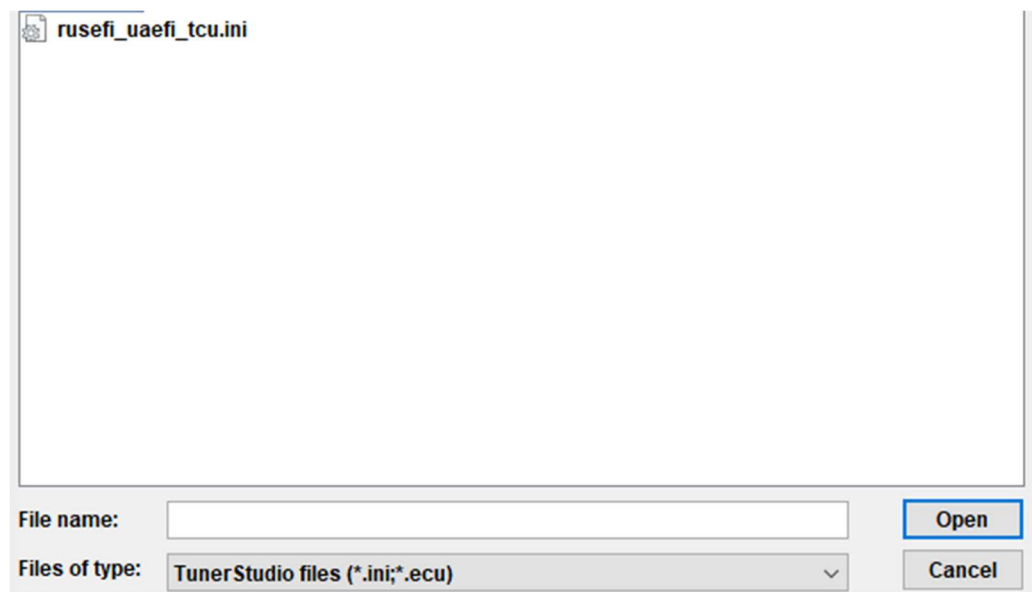
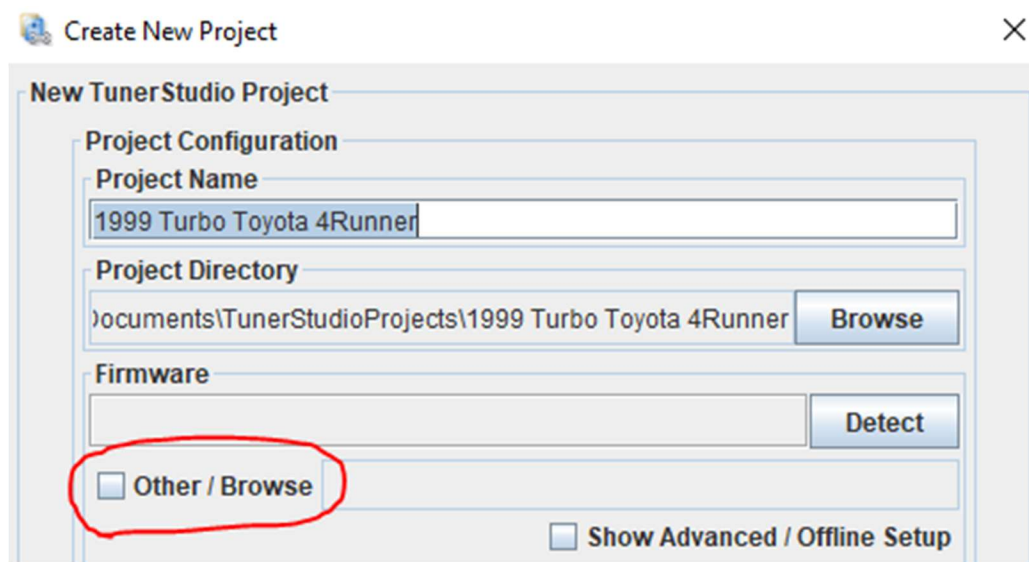
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- h. What happens if TunerStudio fails to detect my firmware?

Reach out using the contact information at the bottom of this guide.

Send a message indicating that you need the .ini file for your ECU.

The .ini file can be sent and loaded manually into TunerStudio by selecting “Other / Browse”





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What is an .ini file?

An .ini file tells TunerStudio how to communicate with your ECU and interpret the data it receives. It defines the available settings, gauges, tables, and menus for the specific firmware version running on your ECU. Because different firmware versions may use different features or memory layouts, TunerStudio requires the correct .ini file to communicate properly.

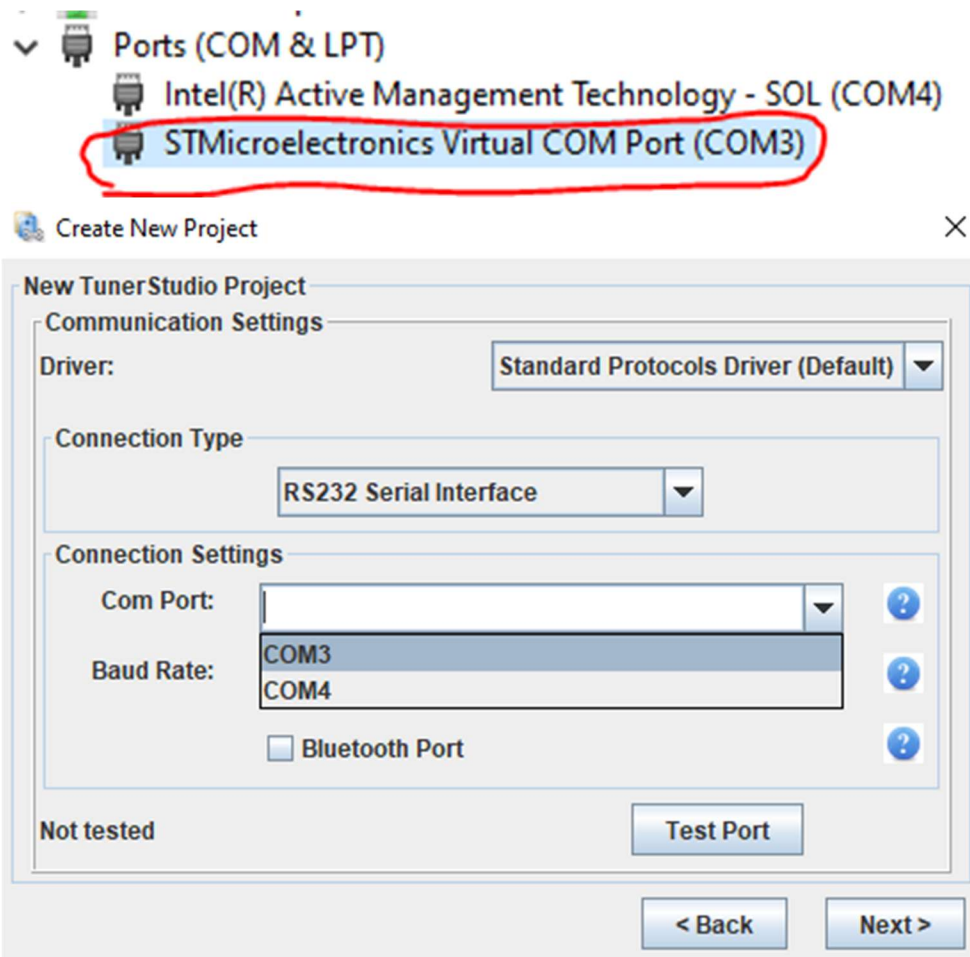
- i. Select “Next”

The screenshot shows the 'Create New Project' dialog box in TunerStudio. The dialog has a title bar with a close button (X) in the top right corner. The main area is titled 'New TunerStudio Project' and contains several sections: 'Project Configuration' with a 'Project Name' field containing '1999 Turbo Toyota 4Runner'; 'Project Directory' with a text field showing a path and a 'Browse' button; 'Firmware' with a text field containing a version string and a 'Detect' button; a checkbox for 'Other / Browse' with an adjacent text field; and a checkbox for 'Show Advanced / Offline Setup'. At the bottom is a 'Project Description' text area. At the very bottom of the dialog are two buttons: 'Cancel' and 'Next >'. The 'Next >' button is circled in red.



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- j. Select the correct port where the ECU is connected to your computer.
To see which port the ECU is connected to, open Device Manager. Under “Ports (COM & LPT)” search for “STMicroelectronics Virtual COM Port” to determine the correct port.



Select the COM port corresponding to “STMicroelectronics Virtual COM Port (COMx)”. The COM number may be different on your computer. Then hit “Next”.

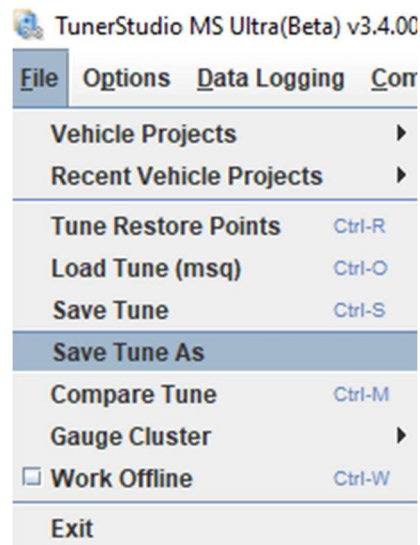


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4. Save your current tune!

Your ECU should now be connected to TunerStudio. Immediately save your current tune before making any changes to it. This is important in case you need to return to the base tune file.

- a. File → Save Tune As...





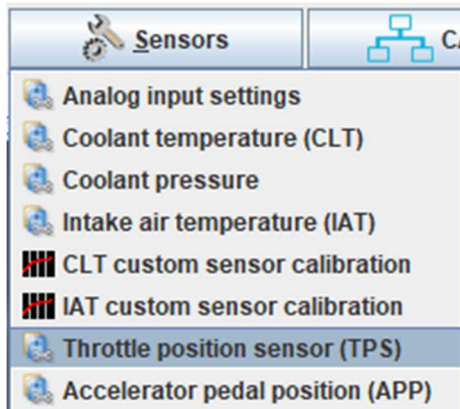
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5. Calibrate your Throttle Position Sensor (TPS)

Once your ECU is connected to Tuner Studio, one of the first things you should do is calibrate your TPS. This is important, particularly for accurate automatic transmission control.

a. Navigate to...

Sensors → Throttle Position Sensor (TPS)





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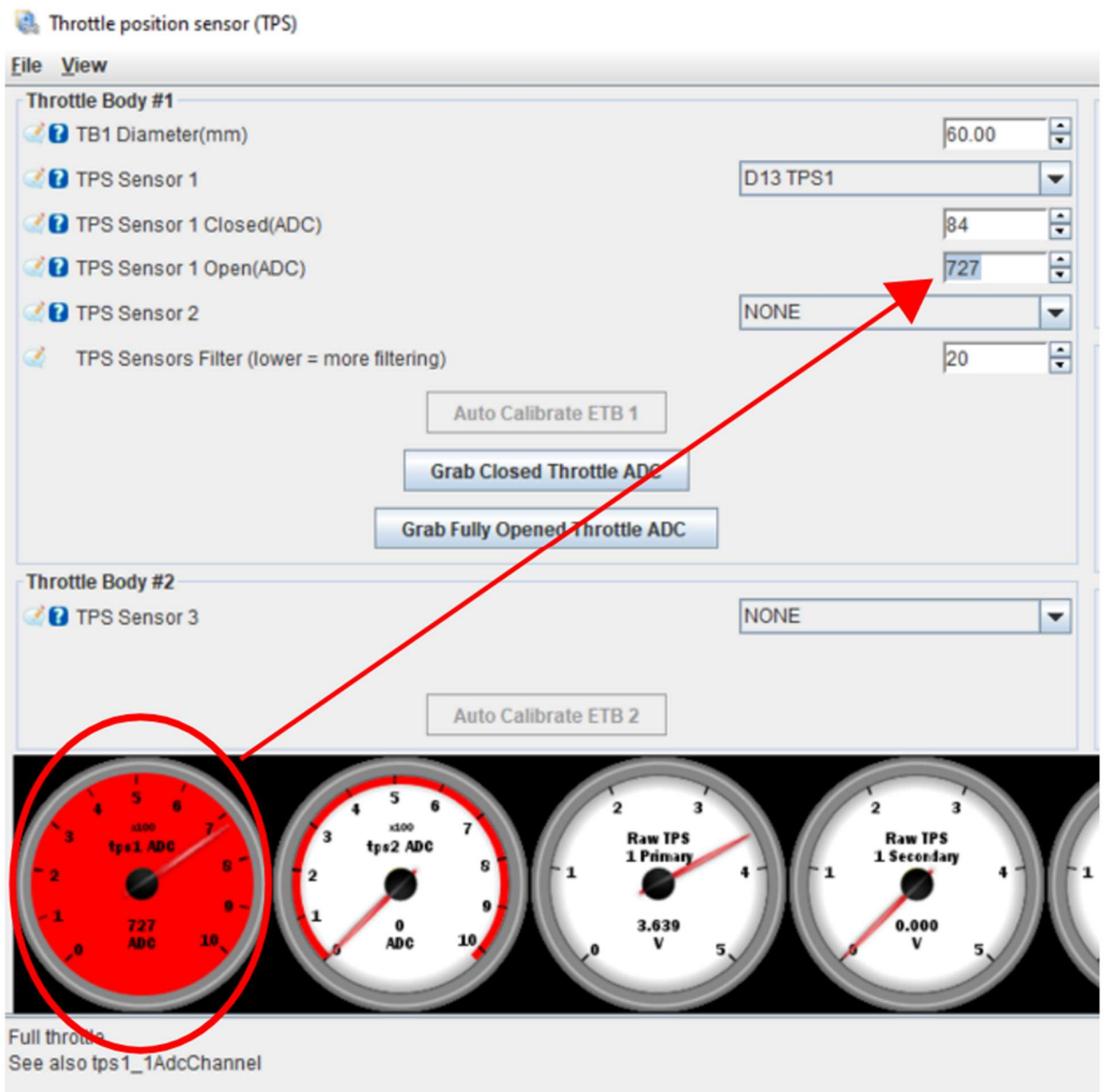
- b. With the throttle completely closed, note the value shown in the ADC gauge.
Enter that value into *TPS Sensor 1 Closed/(ADC)*

The screenshot shows the 'Throttle position sensor (TPS)' configuration window. The 'Throttle Body #1' section is active, showing various settings for TPS Sensor 1. A red arrow points from the 'TPS Sensor 1 Closed(ADC)' gauge in the bottom left to the 'TPS Sensor 1 Closed(ADC)' input field in the top right, which contains the value '84'. The 'TPS Sensor 1 Open(ADC)' field contains '700'. The 'TPS Sensor 2' is set to 'NONE'. The 'TPS Sensors Filter' is set to '20'. Below these are buttons for 'Auto Calibrate ETB 1', 'Grab Closed Throttle ADC', and 'Grab Fully Opened Throttle ADC'. The 'Throttle Body #2' section shows 'TPS Sensor 3' set to 'NONE' and an 'Auto Calibrate ETB 2' button. At the bottom, there are four gauges: 'tp1 ADC' (red face, 0-10 scale, showing 5.4), 'tp2 ADC' (white face, 0-10 scale, showing 0), 'Raw TPS 1 Primary' (white face, 0-5 scale, showing 0.420 V), and 'Raw TPS 1 Secondary' (white face, 0-5 scale, showing 0.000 V). A red circle highlights the 'tp1 ADC' gauge. At the bottom left, text reads: 'Closed throttle, 1 volt = 200 units. See also tps1_adcChannel'.



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- c. With the engine off and throttle wide open, note the value shown in the ADC gauge. Enter that value into *TPS Sensor 1 Open/(ADC)*



- d. Click “Apply” and then “OK” at the bottom of the menu to save changes and exit.



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Need help?

Email:

RMControlSystems@gmail.com

Facebook:

RM Control Systems

<https://www.facebook.com/profile.php?id=61580687264002>

Documentation/Downloads:

<https://github.com/rmcontrolsystems/PNP-Adapter-Boards>